

Integration Coverage

PREPARED BY Energy Hippo implementation team

This matrix tracks what the synthetic customer must eventually prove. It starts as a planning document and should become more concrete as generators and validations are added.

Packet 0.3 supplies the first deterministic AcquiSuite deliveries and the [gateway/format acceptance list](#). All installed UI setup and ingestion acceptance remain **Not run**. Local generator checks establish file contents, not downstream EEMSuite behavior.

Area	Data needed	V1 target	Status
Monthly bill ingestion	Accounts, meters, bills, charges, units, provider metadata	Electric, unmetered lighting, gas, water, sewer, stormwater, irrigation, fire service, chilled water, steam, solar PPA	Planned
Interval ingestion	Channels, readings, units, timestamps, time zone rules	Electric and solar 15-minute; chilled water, Central Plant production and electric submeters hourly	Planned
AcquiSuite gateway	Runtime nodes/points, serial/device/row mappings, AcquiSuite log files	Student Center and Lakeview Residence Hall electric 15-minute kWh and kW; complete, gap and backfill deliveries for Student Center	Generated; installed acceptance pending
NOAA gateway	Runtime station nodes/points, NOAA XML observations, weather type mappings	KSAC temperature, humidity, and wind observations	Planned
Aeris Weather gateway	Runtime station nodes/points, Aeris credentials, forecast/history state	KSMF observed and forecast temperature	Planned
MV90 MDEF gateway	Runtime nodes/points, recorder/channel mappings, .mde files	Science Center, Recreation and Aquatics Center and Central Plant electric 15-minute kWh and kW	Planned
MV90 MV9 gateway	Runtime nodes/points, recorder/channel mappings, .mv9 files	Helios solar generation and export intervals	Planned
BACnet gateway	Runtime nodes/points, device/object mappings, present-value reads	Science Center and Recreation and Aquatics Center chilled-water ton-hours and tons; Central Plant chilled-water supply, steam production and condensate return	Planned
Spinwave gateway	Runtime folder/device/channel mappings and CSV files	Science Center wireless temperature and humidity sensors	Planned
FIG gateway	Point aliases and FIG-family provider file variants	FIG, SIEMANSREPORT, MEDIATOR, EATON, XML, FIXEDNETWORKS, and CMEP aliases for the Cedar Row A water master; a fixed-network submeter file for every Cedar Row apartment and house electric submeter, the Lakeview Residence Hall laundry and the Parking Structure EV chargers	Planned
Modbus gateway	Runtime node TCP settings and register metadata	Thermal plant chiller kW via floating-point holding register	Planned

Area	Data needed	V1 target	Status
ODBC gateway	Source profile, query key, timestamp cursor, token substitution	Library electric interval reads from synthetic SQL historian	Planned
Fake gateway	Generation parameters, state file, optional replay/anomaly settings	Deterministic synthetic interval smoke channel	Planned
Neptune gateway	Handheld upload file, MetaWorldID, usage/register point mappings	Cedar Row B water master route upload; Cedar Row apartment and house water submeter route upload	Planned
MVRS gateway	Handheld upload file, MetaWorldID, usage/register points, event output	Student Center gas route upload plus HMR event records	Planned
HMR events publish	HMR event JSONL, gateway identity, queue/idempotency metadata	MVRS/Neptune event payload publishing	Planned
Onsite generation and attributes	Generation channels, PPA invoice, RECs, avoided CO2e	Helios Onsite Solar generation, net metering, REC statement	Planned
Backfill processes	Date ranges, channel mappings, missing data windows	One missing interval block and one corrected bill	Planned
MDEF generation	Meter definitions, channels, unit metadata, account mappings	Science Center electric interval meter plus provider interval definitions	Planned
BIF generation	Billing account, meter, usage, cost, and charge details	Monthly bills for all provider categories	Planned
Reports	Customer, site, building, meter, provider, bill, and interval rollups	Basic rollups and known totals	Planned
EnergyAI	Building metadata, weather context, interval series, monthly summaries	One campus interval use case	Planned
ENERGY STAR	Property <code>primaryFunction</code> , gross floor area, occupancy, year built, and per-use <code>useDetails</code> ; monthly meter data	Campus parent (College/University) with child building properties, a multifamily property with its buildings, and a standalone Central Plant property — 13 properties mapped in <code>energyStarProperties</code> (see implementation/energystar-integration.pdf)	Planned (property mapping defined)
Weather mapping	Station identifiers, local time zone, fallback station	KSAC primary and KSMF secondary	Planned
Data quality checks	Missing intervals, duplicates, invalid units, corrected bills	Positive and negative scenario assertions	Planned

Coverage Tracking

Use the Status column conservatively:

- **Planned:** scenario is described but no generator or validation exists.
- **Generated:** source data and artifacts can be generated.
- **Load-tested:** artifacts have been loaded into a dev environment.
- **Asserted:** automated validation verifies expected results.

Adding New Coverage

When adding a gateway or integration:

1. Document the minimum required source data.
2. Add or update a scenario in `implementation/scenario-catalog.pdf`.
3. Add fields to the source model only when they represent durable domain data.
4. Generate artifacts from the source model.
5. Add assertions that prove the downstream behavior.